

WEST HAWAII BUOY 51003, USA

WMO: 992610

Lat: 19.175N

Lon: 160.625W

Elev: 0

StdP: 14.70

Time Zone: -11.00 (W11)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	71.0	71.9	N/A	N/A	N/A	N/A	N/A	N/A	30.6	73.7	26.9	73.9	15.6	40	0.563

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	2.7	82.8	N/A	82.3	N/A	81.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	11.6	70

Dehumidification DP/MCDB and HR

0.4%			1%			2%			0.4%		1%		2%		Exterior Max WB
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
24.8	23.0	21.6	68.3	83.2	1.4	1.0	67.3	83.9	66.5	84.4	65.7	85.0	64.6	85.7		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	77.2	75.2	75.0	74.7	75.4	76.4	77.7	78.6	79.5	79.6	79.4	78.1	76.6
	DBStd	2.21	1.38	1.40	1.32	1.33	1.28	1.15	1.21	1.15	1.31	1.20	1.30	1.40
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	9923	780	699	766	761	819	830	886	914	887	913	844	824
	CDD65	4448	315	279	301	311	354	380	421	449	437	448	394	359
	CDH74	31467	1172	949	937	1335	2062	2916	3685	4318	4307	4346	3253	2187
	CDH80	1095	0	0	0	0	1	7	81	269	390	295	51	1
Wind	WSAvg	13.9	13.3	14.0	15.4	15.2	13.3	13.3	13.8	13.4	12.2	13.2	15.2	15.0
Precipitation	PrecAvg	17.2	1.5	2.0	2.0	1.0	0.6	0.5	0.7	1.1	1.1	2.0	2.4	2.4
	PrecMax	34.6	6.3	6.0	9.1	4.9	2.6	3.7	2.2	5.0	5.4	5.1	8.4	10.8
	PrecMin	8.0	0.2	0.1	0.1	0.2	0.2	0.0	0.1	0.2	0.1	0.2	0.1	0.2
	PrecStd	6.6	1.4	1.9	2.2	1.0	0.6	0.6	0.5	1.1	1.1	1.3	2.0	2.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.3	78.5	78.3	78.5	80.3	80.8	82.3	83.4	84.2	82.5	81.7	80.3
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	78.4	78.2	77.5	78.3	79.6	80.4	81.5	82.4	83.3	82.3	81.2	79.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	77.7	77.7	77.3	78.0	79.1	80.1	81.0	81.9	82.4	81.9	80.6	79.2
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	77.3	77.3	77.0	77.5	78.4	79.6	80.4	81.3	81.9	81.4	80.3	78.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	2.6	2.6	2.7	2.6	2.5	2.6	2.5	2.4	2.7	2.5	2.6	2.4
		MCDBR	3.0	2.8	2.8	2.6	2.6	2.7	2.4	2.5	3.1	2.6	2.8	2.6
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.360	0.364	0.375	0.410	0.407	0.398	0.400	0.398	0.387	0.381	0.374	0.361	
	taud	2.601	2.588	2.562	2.436	2.468	2.492	2.501	2.506	2.525	2.590	2.595	2.616	
	Ebn at Noon	293	297	296	285	282	282	282	284	288	288	286	289	
	Edh at Noon	29	31	33	37	36	35	35	35	33	30	29	28	
All-Sky Solar Radiation	RadAvg	1505	1656	1837	2009	2120	2141	2139	2099	1966	1700	1462	1377	
	RadStd	65	97	188	142	130	80	62	67	80	69	60	89	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page